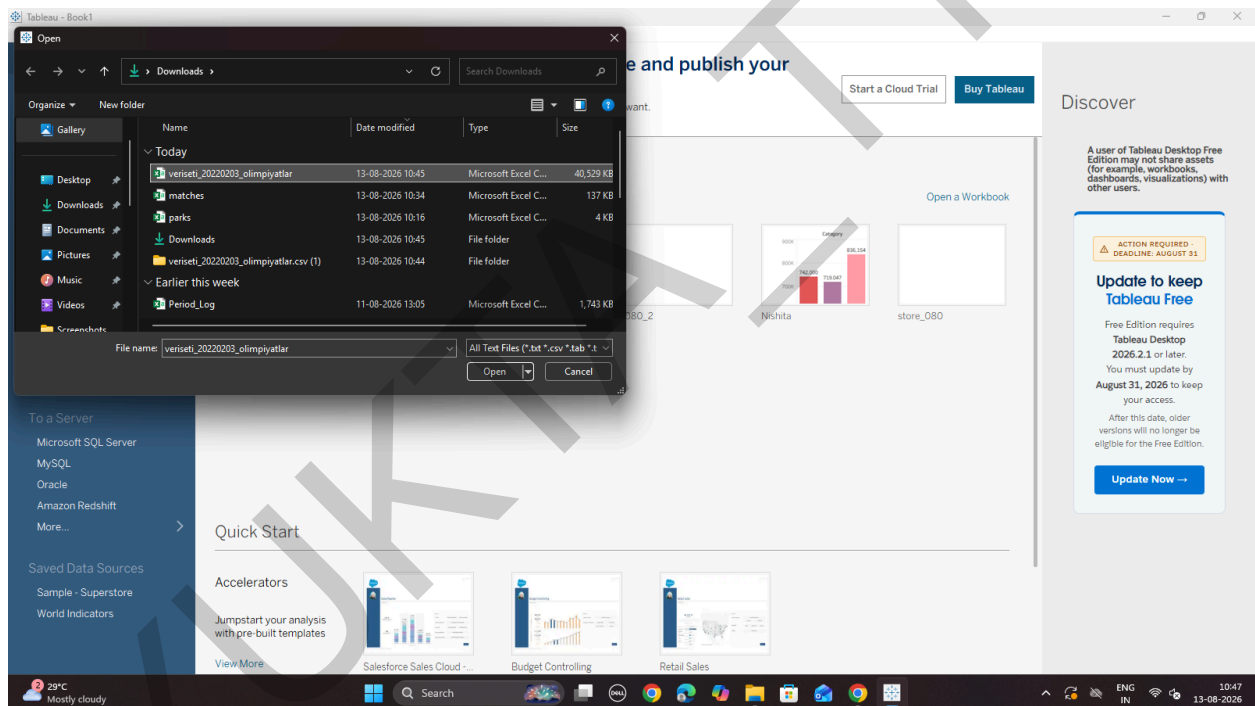


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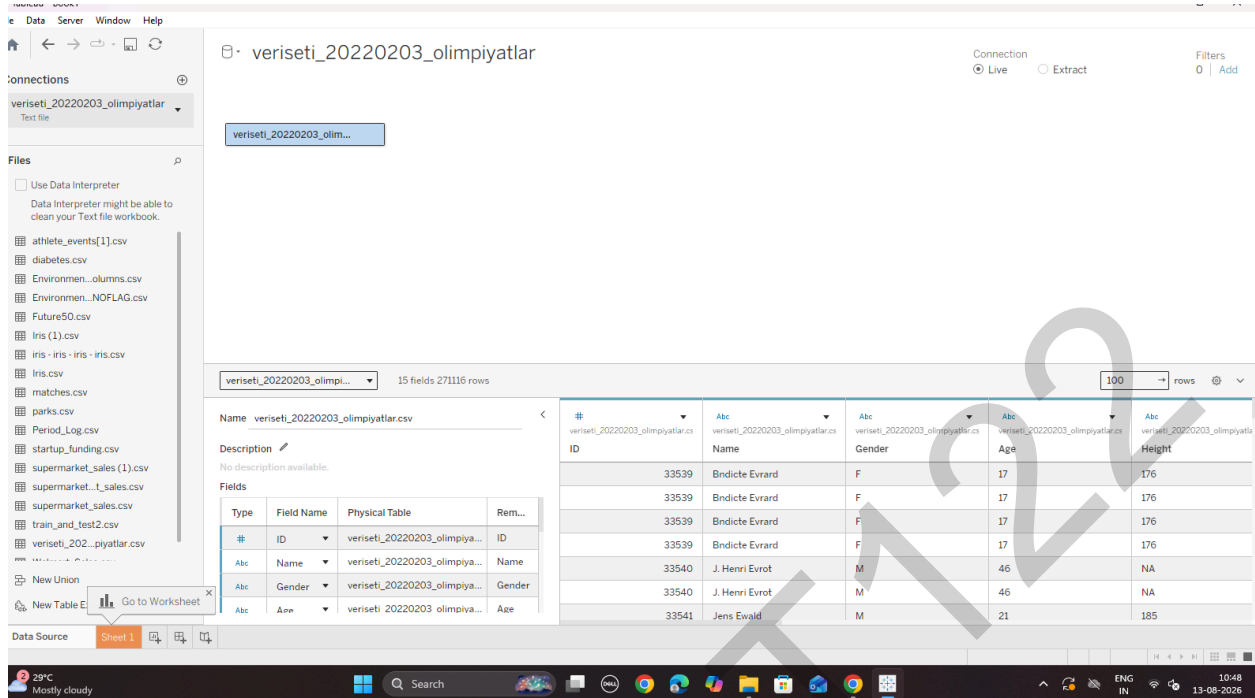
Aim : Practical 7: Creating Interactive Dashboards and Data Stories

1. Connect the Dataset to Tableau

1. Open **Tableau Desktop**.
2. Select **File** → **Open** or **Connect to Data**.
3. Choose **Text File**.
4. Select:
verisetti_20220203_olimpiyatlar.csv
5. Tableau will open the **Data Source** page.
6. Check that the fields are correctly recognized.
7. Click **Sheet 1** to start creating visualizations.



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2. Create Basic Sheets

Create several worksheets first. These sheets will later be combined into a dashboard.

Sheet 1 – Total Athletes

1. Rename the sheet to **Total Athletes**.
2. Drag **Name** to **Text**.
3. Right-click **Name**.
4. Select **Measure** → **Count (Distinct)**.
5. You will get the total number of unique athletes.

Purpose: Shows the total number of athletes in the dataset.

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The image displays two screenshots of the Tableau Desktop interface, illustrating the steps to create a 'Total Athletes' visualization.

Top Screenshot: The Tableau interface shows a worksheet named 'Total Athletes'. The data source is 'veriseti_20220203_olim...'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic'. A context menu is open for the 'Name' field, showing options like 'Filter...', 'Show Filter', 'Show Highlighter', 'Sort...', 'Format...', 'Include in Tooltip', 'Edit Aliases...', 'Find in Data pane', 'Dimension', 'Attribute', 'Measure', 'Measure in Shelf', and 'Remove'. The 'Measure' option is selected, and a sub-menu is open showing 'Minimum', 'Maximum', 'Count', and 'Count (Distinct)'. The 'Count (Distinct)' option is highlighted.

Bottom Screenshot: The Tableau interface shows the same worksheet. The 'Name' field is now on the 'Columns' shelf. The 'Marks' card is set to 'Automatic'. The 'Total Athletes' visualization is displayed as a single cell with the value '1,34,732'. The 'Columns' shelf now contains 'CNTD(Name)'. The 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic'.

Sheet 2 – Athletes by Year

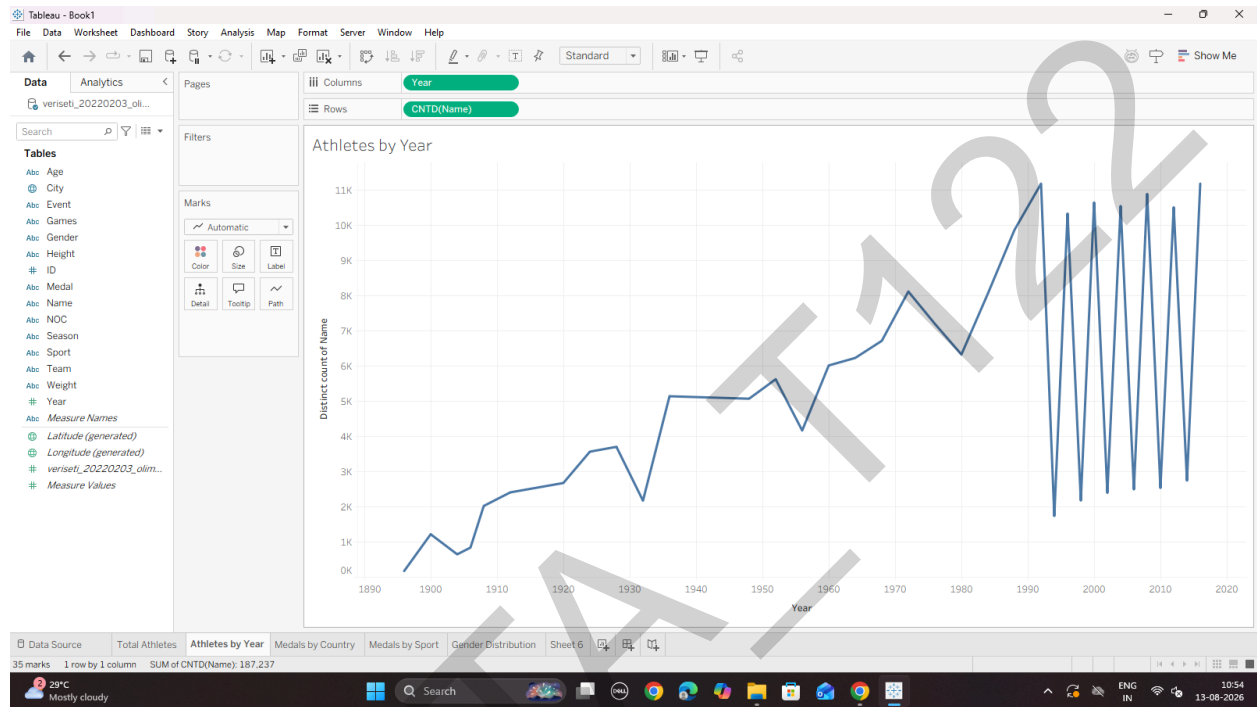
1. Create a new worksheet.
2. Rename it **Athletes by Year**.
3. Drag **Year** to **Columns**.

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4. Drag **Name** to **Rows**.
5. Change **Name** to **Count Distinct**.
6. Select the **Line Chart** from Show Me.

Purpose: Shows how athlete participation changes over the Olympic years.

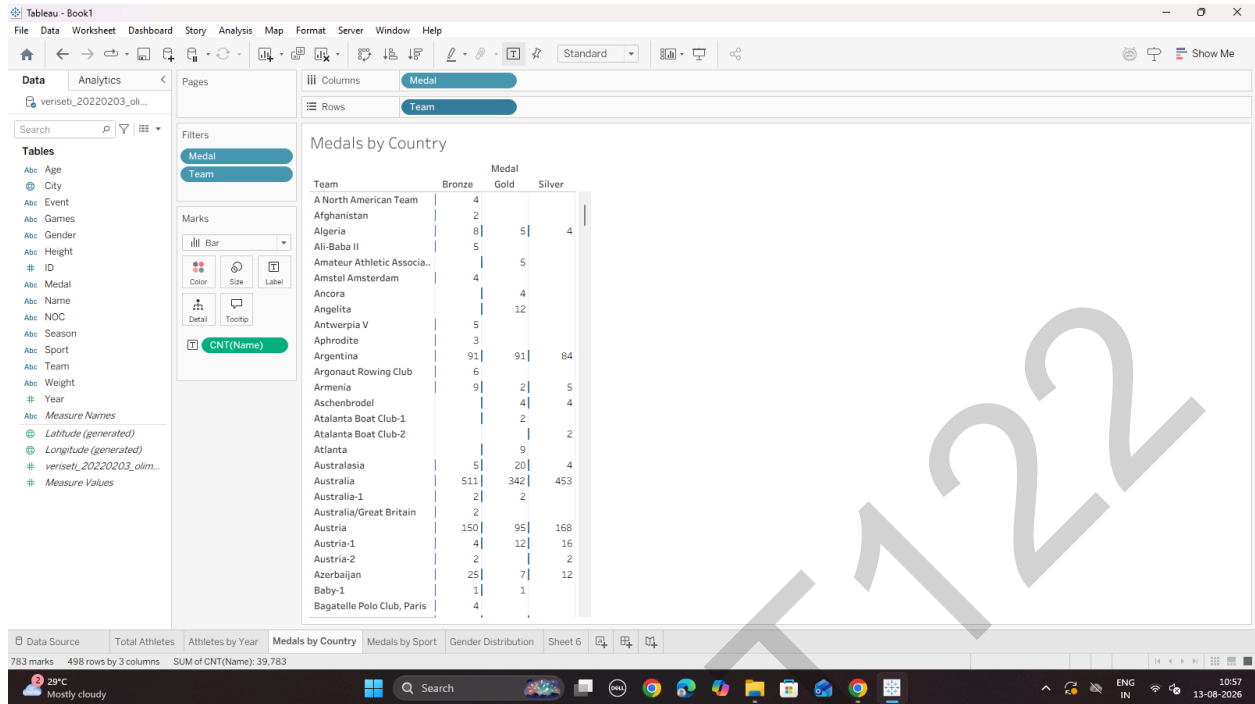


Sheet 3 – Medals by Country

1. Create a new worksheet.
2. Rename it **Medals by Country**.
3. Drag **Team** to **Rows**.
4. Drag **Medal** to **Columns**.
5. Drag **Name** to **Text**.
6. Change **Name** to **Count**.
7. Filter **Medal**.
8. Select:
 - Gold
 - Silver
 - Bronze
9. Choose **Bar Chart**.

Purpose: Compares Olympic medal counts among countries.

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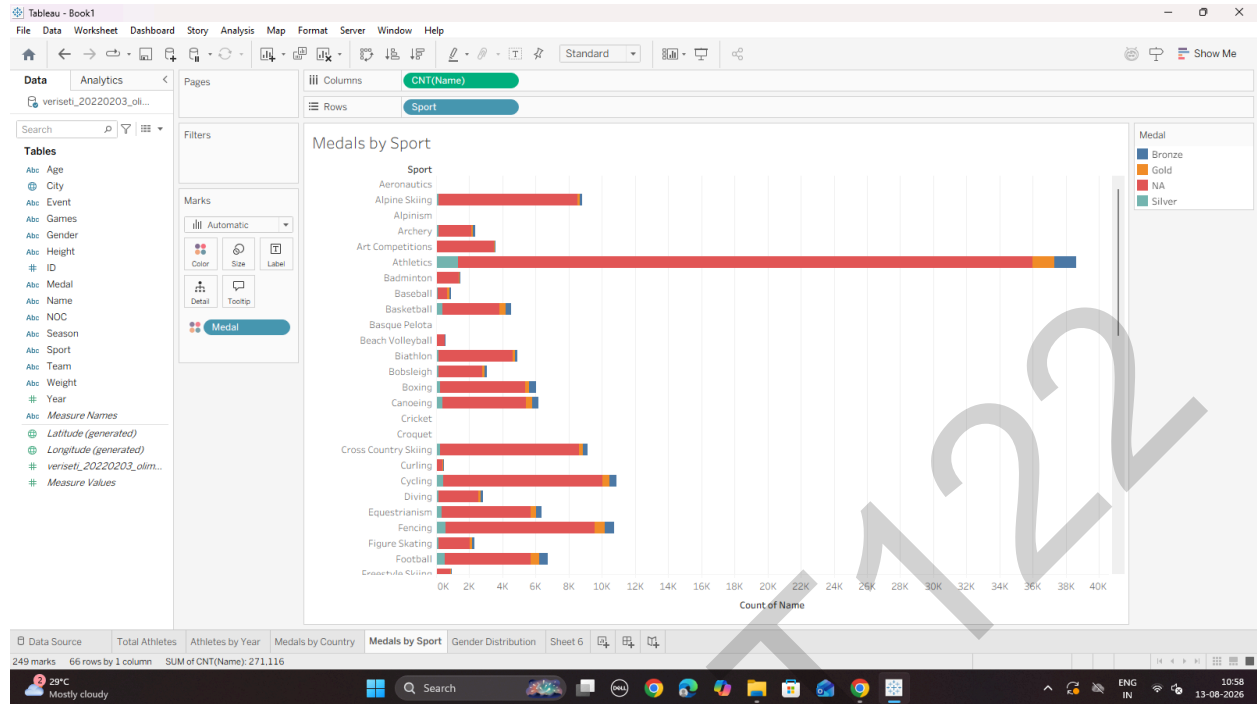


Sheet 4 – Medals by Sport

1. Create a new worksheet.
2. Rename it **Medals by Sport**.
3. Drag **Sport** to **Rows**.
4. Drag **Name** to **Columns**.
5. Change **Name** to **Count**.
6. Drag **Medal** to **Color**.
7. Filter out blank Medal values.

Purpose: Shows which sports have the highest number of medals.

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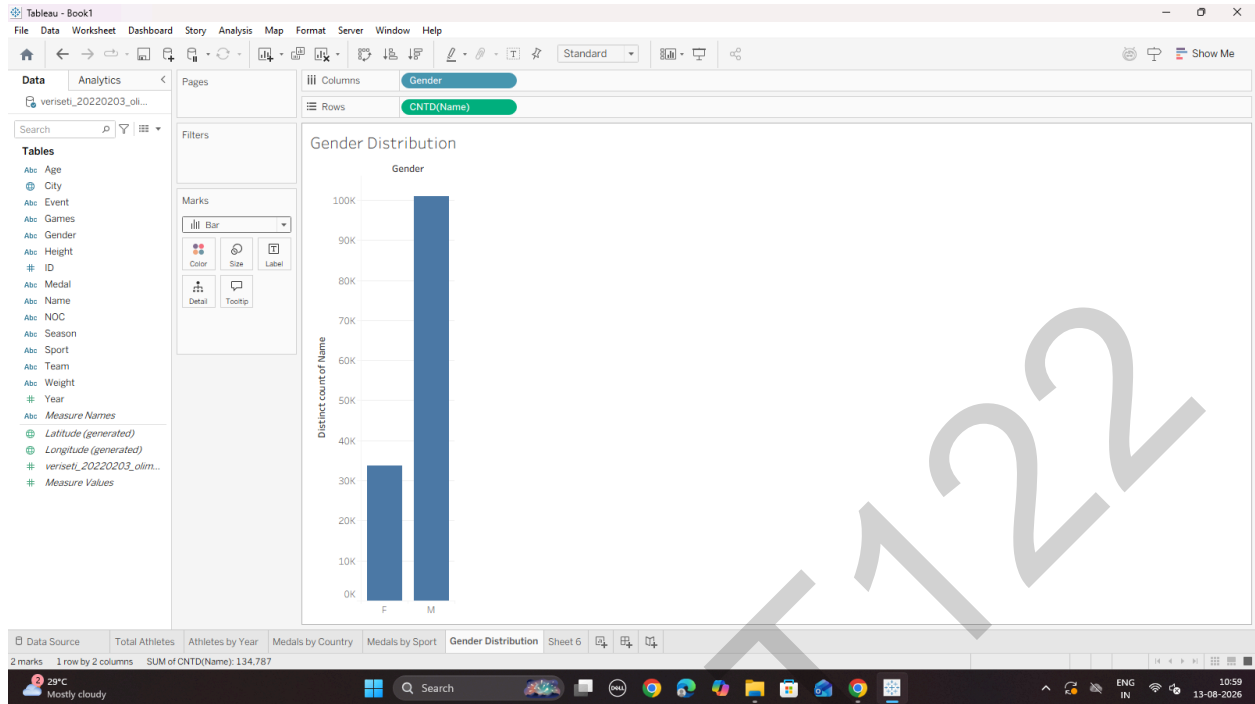


Sheet 5 – Gender Distribution

1. Create a new worksheet.
2. Rename it **Gender Distribution**.
3. Drag **Gender** to **Columns**.
4. Drag **Name** to **Rows**.
5. Change **Name** to **Count Distinct**.
6. Select a **Bar Chart**.

Purpose: Compares male and female athlete participation.

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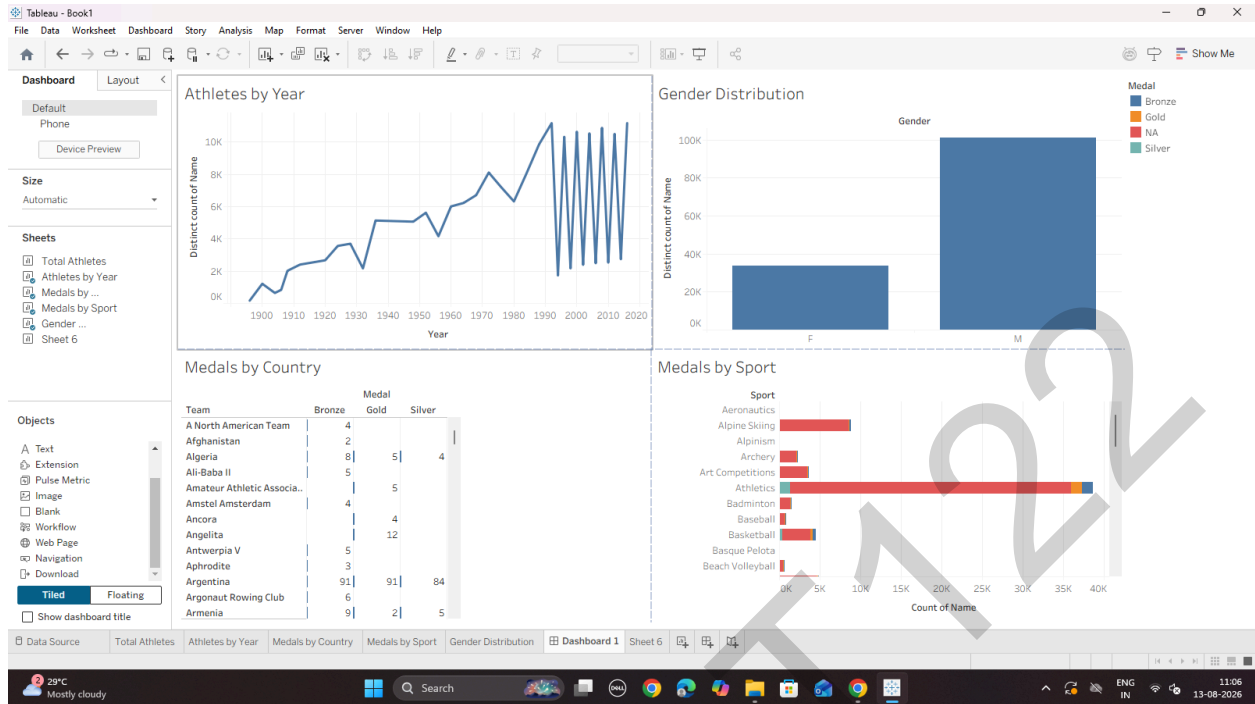


3. Create the Basic Dashboard

Now combine the worksheets.

1. Click **New Dashboard** at the bottom.
2. Set the dashboard size to **Automatic**.
3. Drag **Athletes by Year** onto the dashboard.
4. Drag **Medals by Country** below or beside it.
5. Drag **Medals by Sport** onto the dashboard.
6. Add **Gender Distribution**.
7. Add the **Total Athletes** sheet as a KPI card.

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4. Add Filters

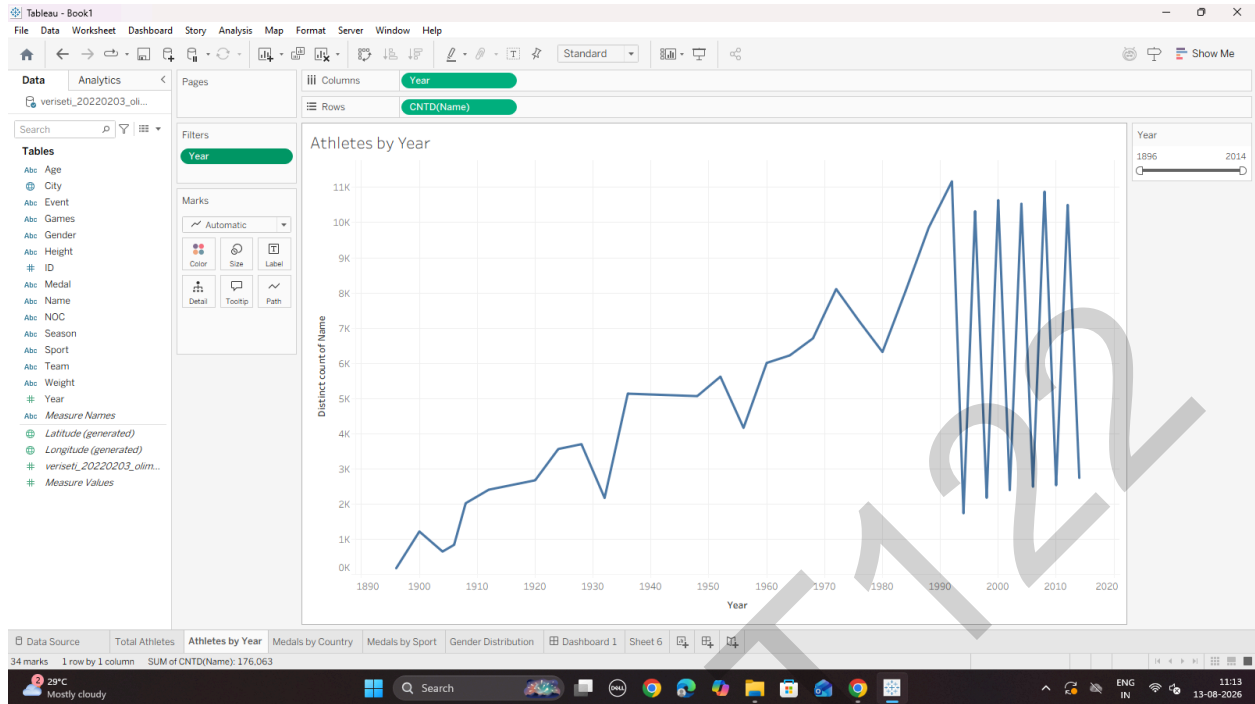
Filters allow users to interactively change the dashboard.

Add Year Filter

1. Open the **Athletes by Year** worksheet.
2. Drag **Year** to **Filters**.
3. Select **All values**.
4. Right-click the filter.
5. Select **Show Filter**.

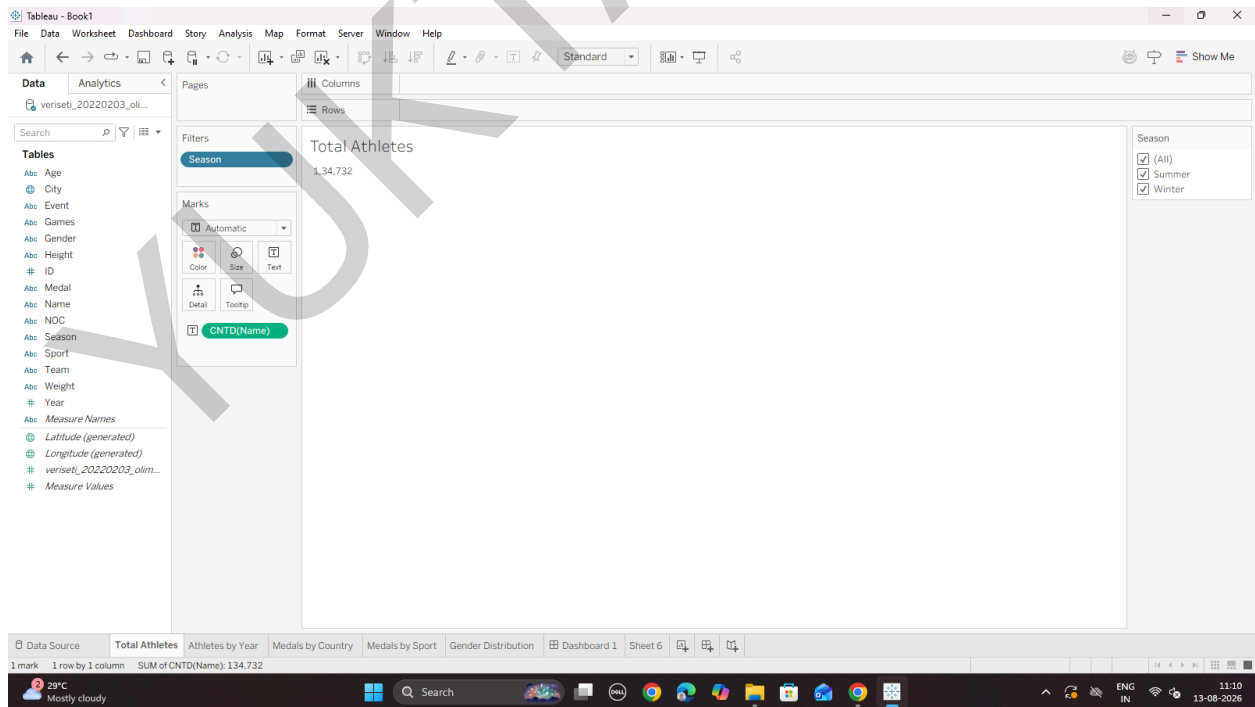
You can now select individual Olympic years.

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Add Season Filter

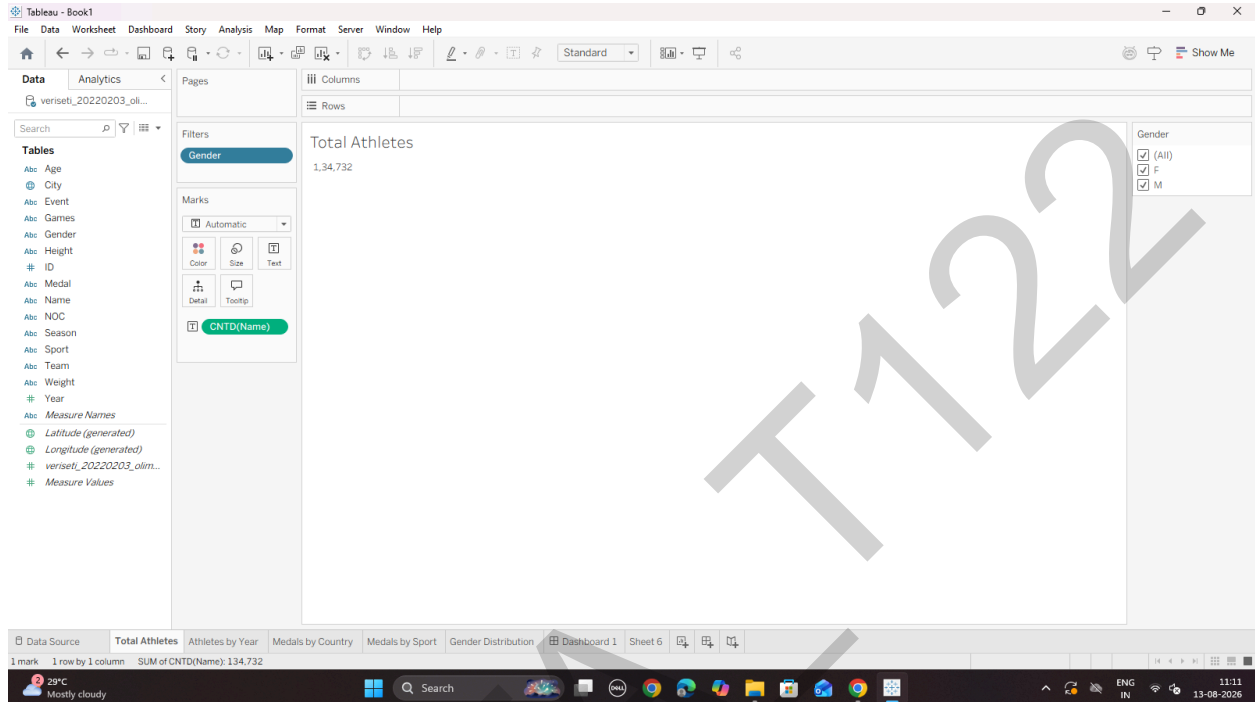
1. Drag **Season** to Filters.
2. Select **All**.
3. Select **Show Filter**.



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Add Gender Filter

1. Drag **Gender** to **Filters**.
2. Select **Show Filter**.



5. Add Interactive Controls

Interactive controls allow users to explore the dashboard.

Useful controls for this dataset are:

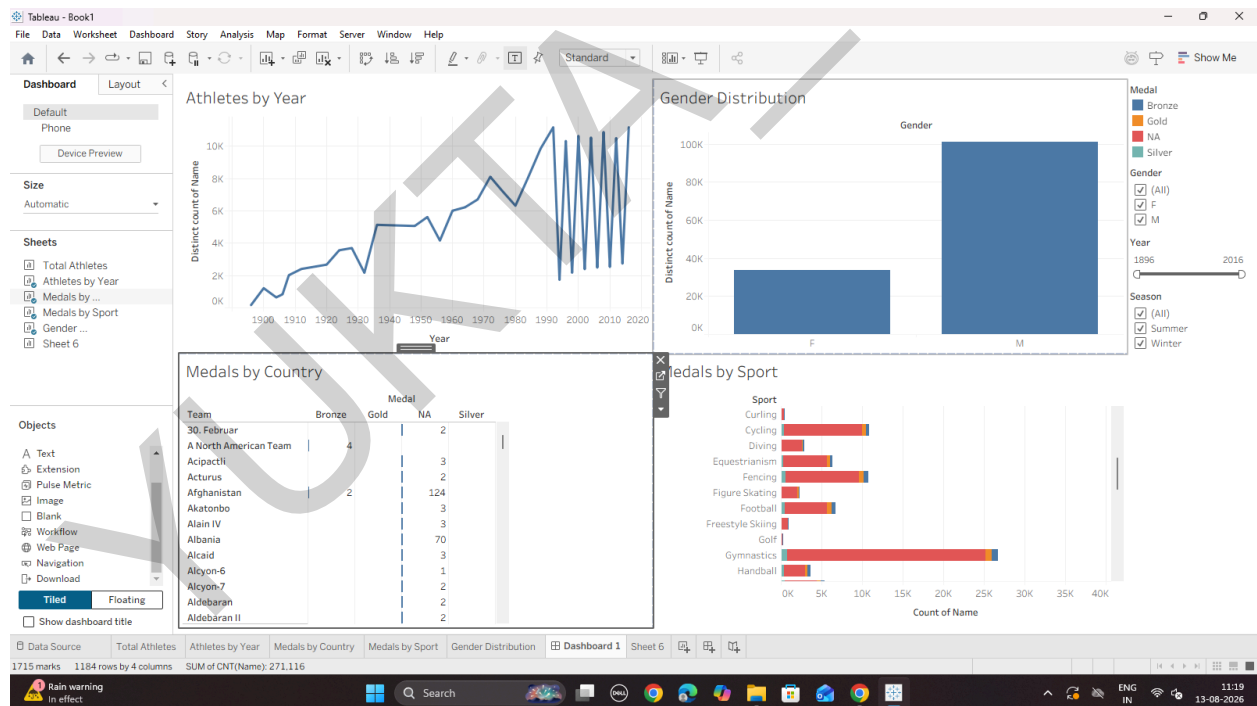
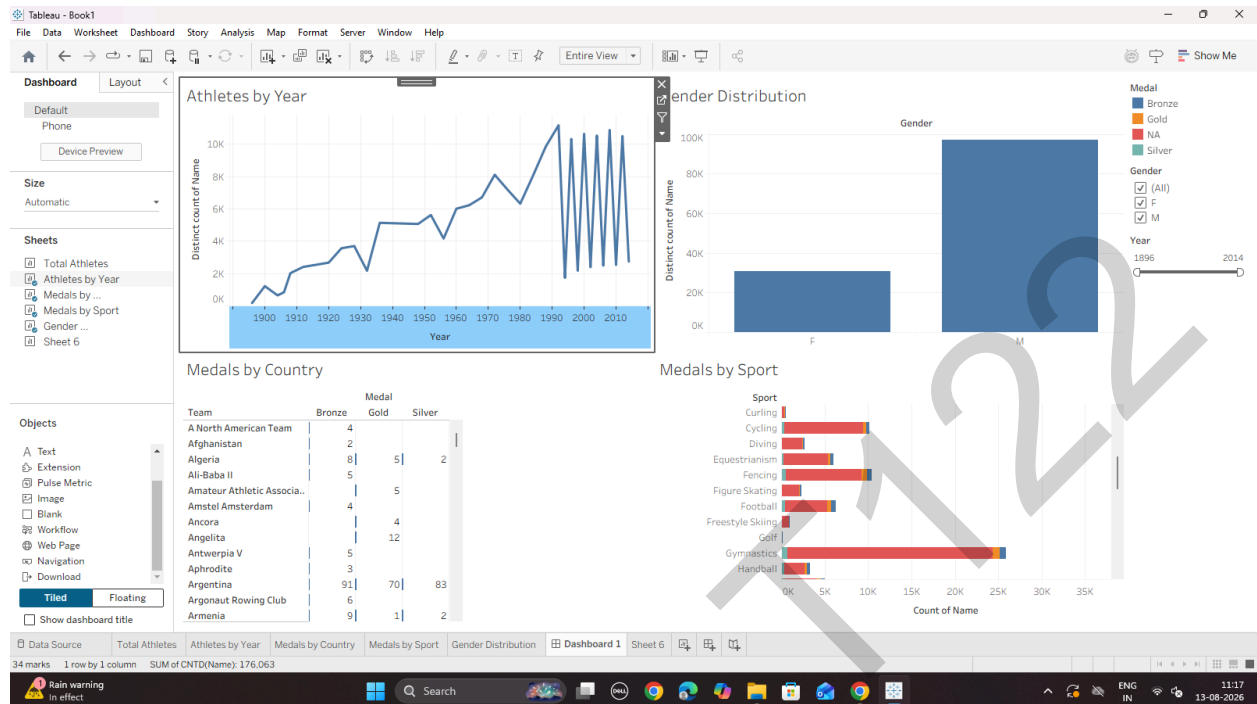
- Year
- Season
- Gender
- Sport
- Team
- Medal

For example:

1. Select the **Year** filter.
2. Change its style to **Single Value Dropdown**.
3. Select **2020**.
4. The dashboard will update to show information for 2020.

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You can similarly use **Sport** as a dropdown.



6. Implement Highlight Actions

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Highlight actions emphasize related data when the user selects a particular value.

Example: Highlight a Sport

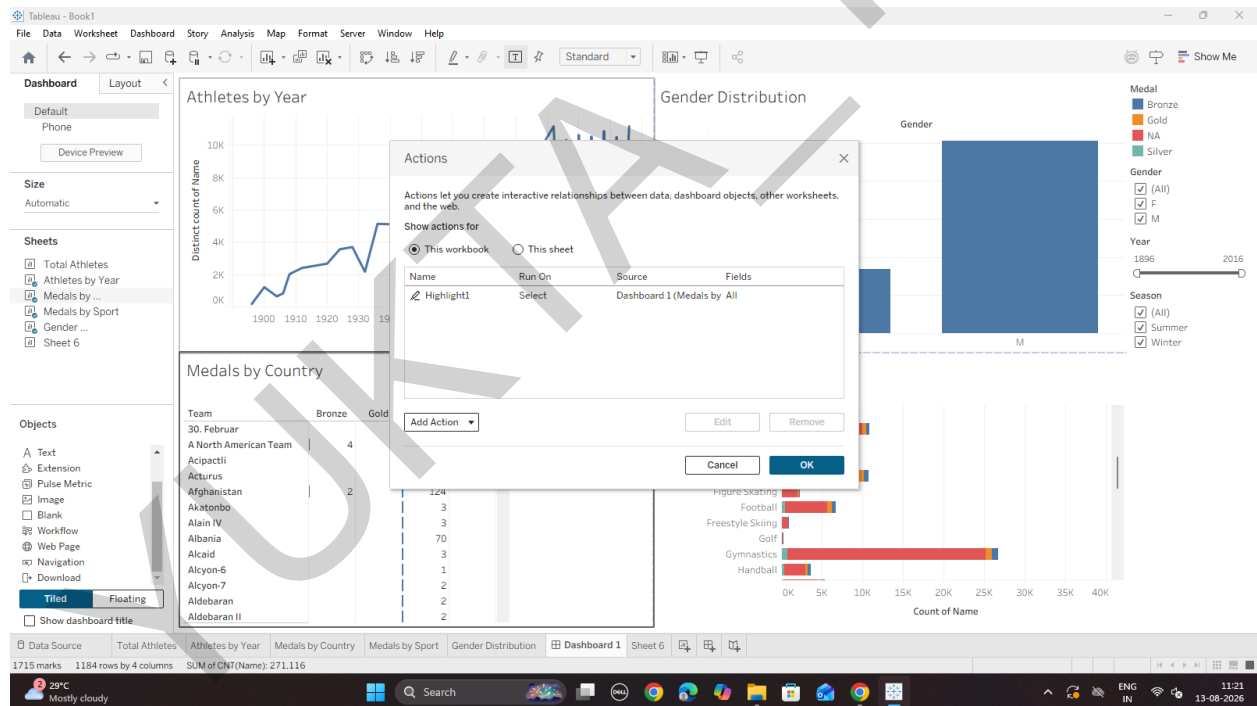
1. Open the dashboard.
2. Select **Dashboard** → **Actions**.
3. Click **Add Action** → **Highlight**.
4. Set:

Source Sheet: Medals by Sport

Target Sheet: Medals by Country

5. Select **Run action on: Select**.
6. Click **OK**.

Now, when you select a sport, related information will be highlighted in other charts.



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7. Implement Filter Actions

A filter action allows one chart to filter another chart.

Example

When the user clicks a country, display only that country's sports and medals.

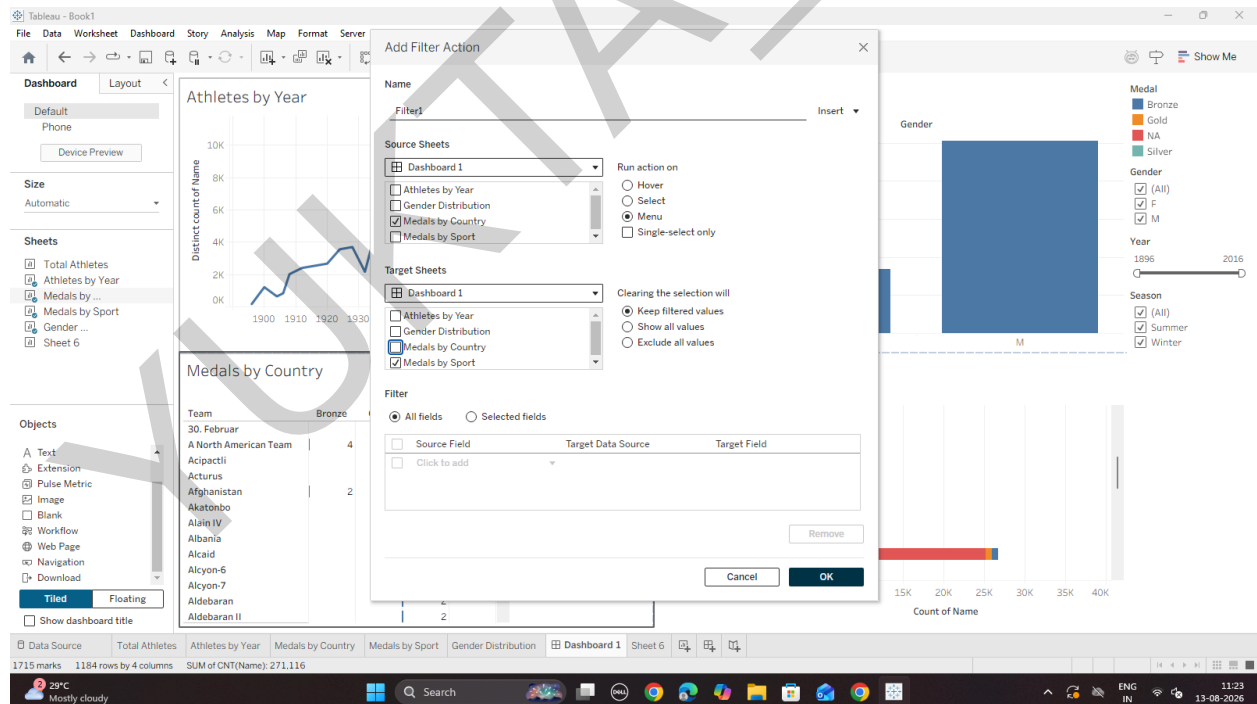
1. Open the dashboard.
2. Select **Dashboard** → **Actions**.
3. Select **Add Action** → **Filter**.
4. Select:

Source: Medals by Country

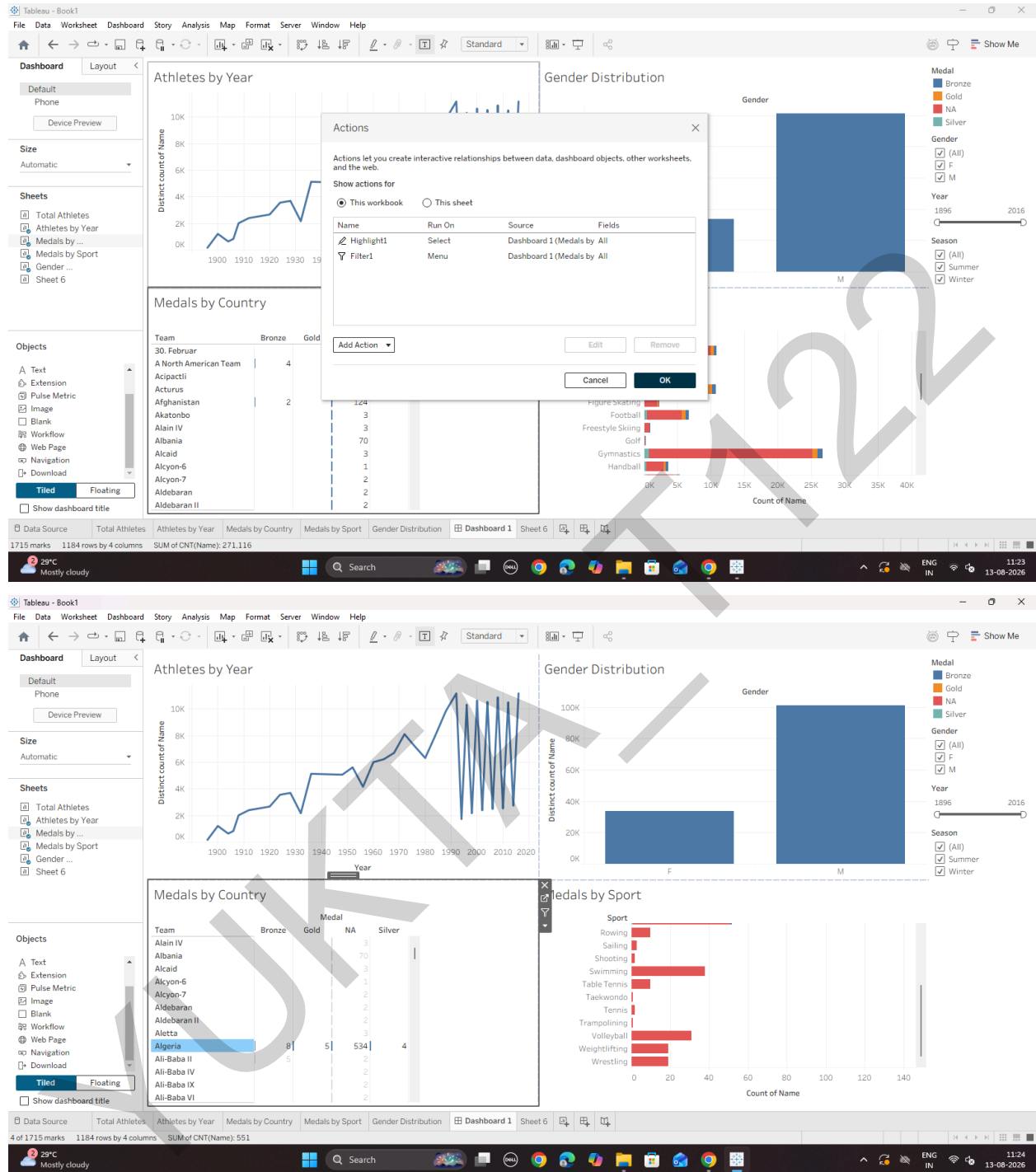
Target: Medals by Sport

5. Select **Run action on** → **Select**.
6. Click **OK**.

Now clicking a country will filter the sport chart.



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8. Use URL Actions

URL actions can open an external webpage when a user selects an item.

For example, you can create a link to Wikipedia for an Olympic sport or country.

Steps

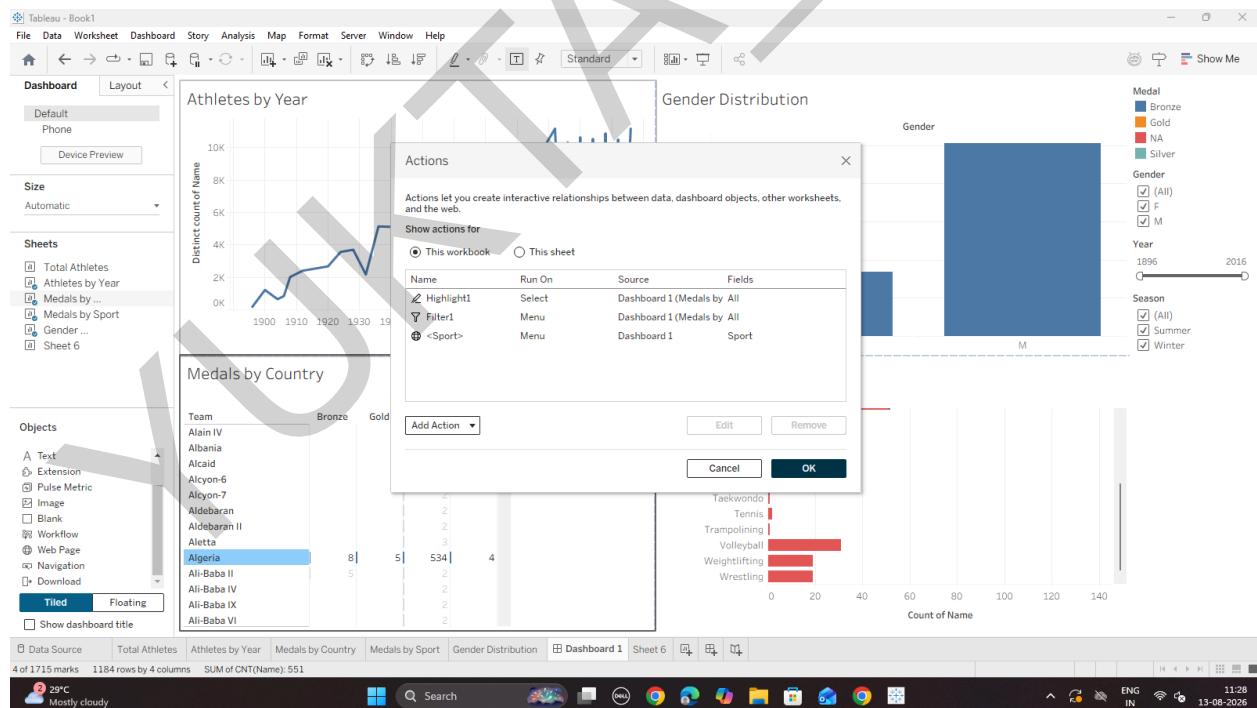
1. Open the dashboard.
2. Go to **Dashboard** → **Actions**.
3. Click **Add Action** → **Go to URL**.
4. Enter a URL such as:

<https://en.wikipedia.org/wiki/<Sport>>

5. Use the **Insert** option to insert the **Sport** field dynamically if supported by the URL action.
6. Set **Run action on** → **Select**.
7. Click **OK**.

When a user selects a sport, Tableau can open the corresponding webpage.

Purpose: Provides additional external information about selected Olympic data.



9. Use Set Actions

Set actions allow users to dynamically select members of a dimension.

Example: Select Countries

First create a set.

1. Open the **Medals by Country** worksheet.
2. Right-click **Team**.
3. Select **Create** → **Set**.
4. Name it:

Selected Countries

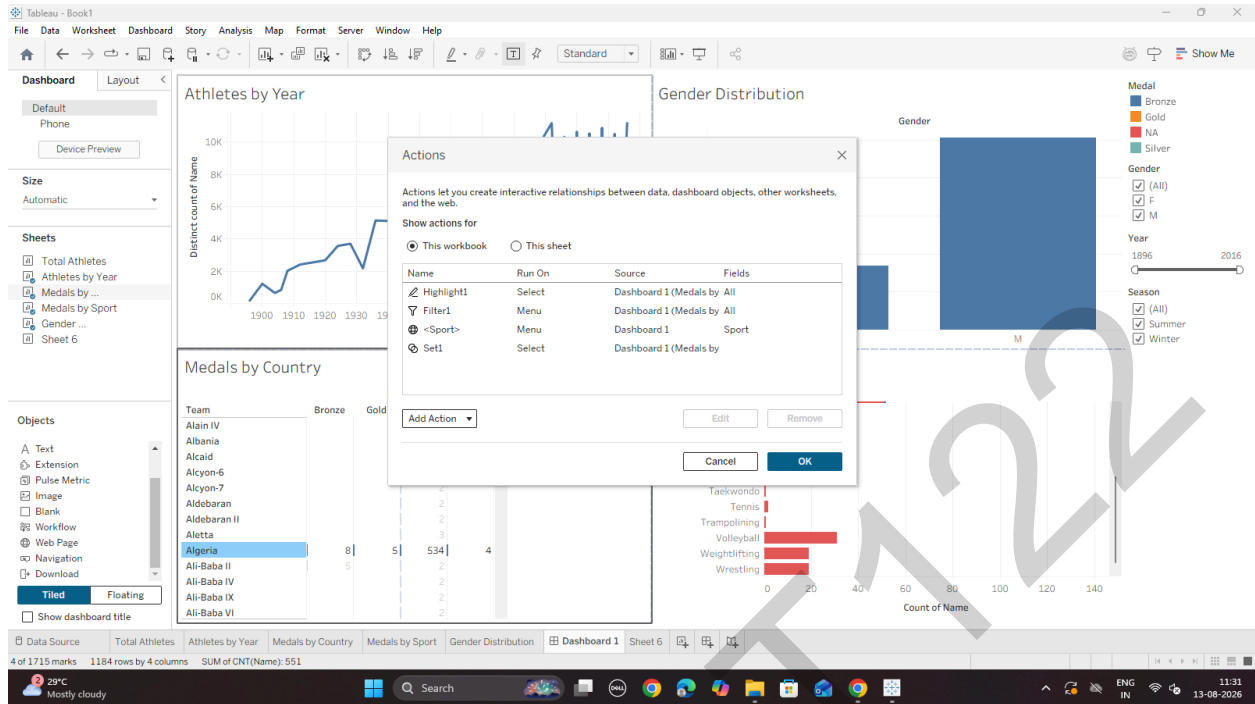
5. Leave the set initially empty or select some countries.
6. Click **OK**.

Now create the action:

1. Open the dashboard.
2. Select **Dashboard** → **Actions**.
3. Click **Add Action** → **Change Set Values**.
4. Select:
 - Source Sheet: **Medals by Country**
 - Target Set: **Selected Countries**
5. Select **Run action on** → **Select**.
6. Click **OK**.

Now clicking countries dynamically changes the set.

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10. Create a Dynamic Dashboard Using Parameters

Parameters allow users to dynamically select what they want to analyze.

For example, create a parameter to select between:

- Athletes
- Medals
- Average Age

Create Parameter

1. Right-click in the **Data pane**.
2. Select **Create Parameter**.
3. Name it:

Select Measure

4. Set **Data Type** → **String**.
5. Select **List**.
6. Add:

Athletes

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Medals

Average Age

7. Click OK.

